REDUCTION OF THE MINIMAL-101 ROOT-4 ODD PERFECT NUMBER BRANCH TO THE SINGLE FACTOR $\Phi_{31}(269)$

ARGUS OPEN RESEARCH PROJECT

Abstract. In the factor-chain tree search for an odd perfect number $N = q^k m^2$ on the minimal-101, special-prime-5, α_1-root-4 branch, we replay 6,756,355 strict-suffix prefixes from a certified 2,054-entry factor database and prove that every branch terminates except at the single mandatory cyclotomic value $\Phi_{31}(269) = \sigma(269^{30}) = (269^{31} - 1)/268$, a 73-digit/96-digit cofactor input outside the pre-certified factor table.

1. Main Factor-Chain Reduction

Proposition 1.1. *Starting from the certified* 2,054*-entry prime-power divisor-sum database on the root-*4 *branch, exhaustive strict-suffix propagation across* 6,756,355 *prefixes closes every divisor-sum branch save for the single unfactored cyclotomic value* $\Phi_{31}(269) = \sigma(269^{30})$.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 8 in the domain of elementary and computational number theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped branch reduction* deciding the scope: 6,756,355 strict-suffix prefixes from 2,054 certified factor entries reduce the root-4 branch to Phi_31(269) = sigma(269^{}30).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- nonexistence of odd perfect numbers.

References

[1] Primary problem list and source literature for Result 8 (elementary and computational number theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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